

Paths – Group Sheet

Math 252 – Spring 2020

Instructions

- A. Work in groups of 3 or 4. Put all of your group members' names in the space below:

 - B. Have one person in your group be the official recorder for the group. The recorder's job is to record your work on some of the problems on this sheet. Feel free to rotate the recorder duties. You should work on all of the problems, but you only have to turn in the ones on this sheet. Turn in as much as you finish during class.
 - C. At the end of the hour, make sure all of these group sheets get turned in.
 - D. The homework due today will be collected on Friday instead of today.
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Paths and Isomorphism

- 4. Consider the pair of graphs below. Are the graphs isomorphic? Explain.

Can you trace it?

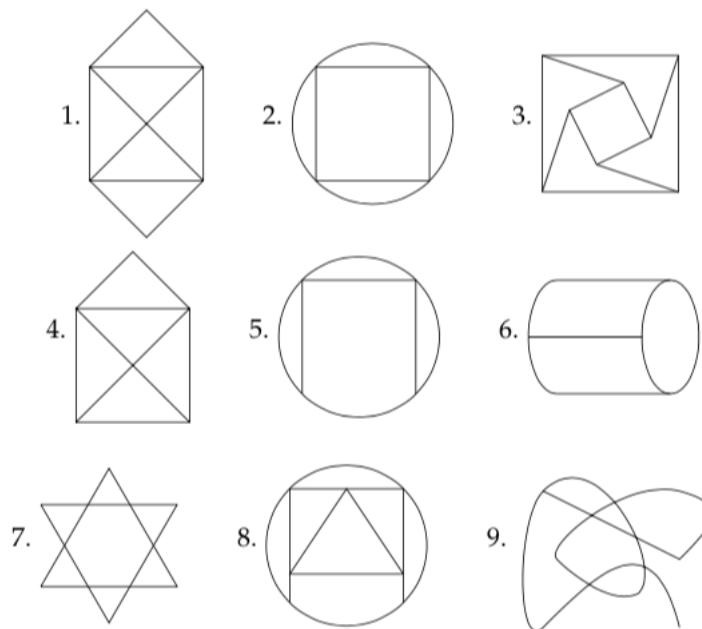
- 6. Which kind of path is this puzzle asking for?

Relationships in Graphs

- 8. Show that K_4 is a planar graph by drawing it without any edge crossings.

- 9. Show that Q_3 is a planar graphs by drawing it without any edge crossings.

10. Add dots showing vertices so that each graph is a planar graph.



11. Fill in the table below for the graphs above.

Graph	# even vertices	# odd vertices	total degree	number of vertices	number of edges	number of regions	Euler path?	Euler circuit?
1								
2								
3								
4								
5								
6								
7								
8								
9								